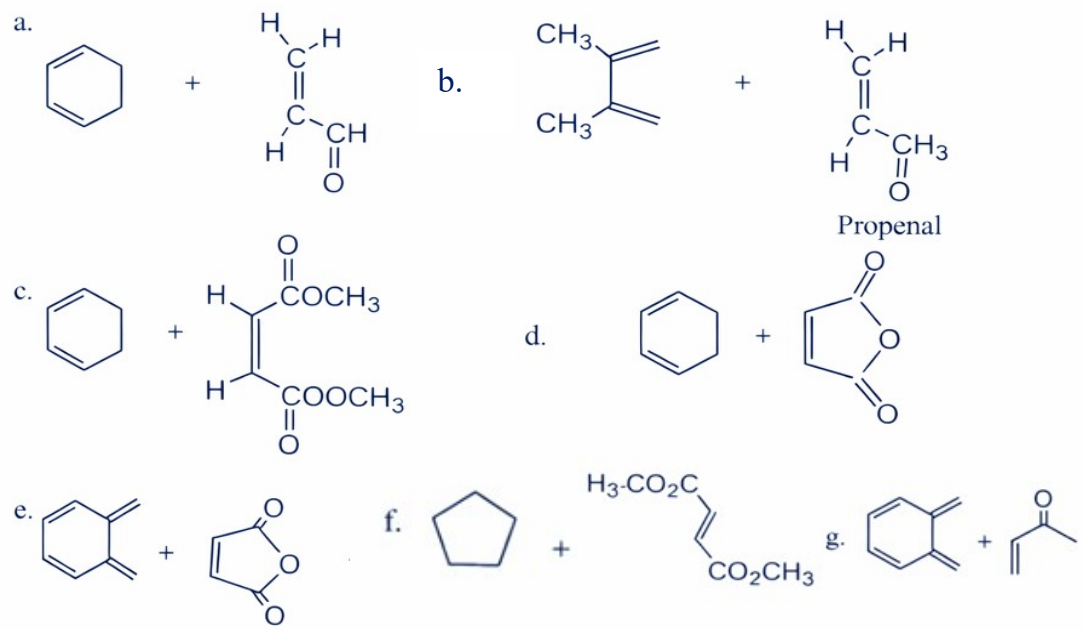
Name ……
Department ……
Reg. No. ……

All Exercises

1. Show how

   (a) barbituric acid could be prepared from urea (b) Show the reactions of urea with malonic esterisocyanate from methyl amine (d) preparation of 2, 3-dimethyl butanoic acid (e) phosgene from ammonia, ethyl alcohol.

2. What is active methylene group? How can you prepare butanoic acid from ethyl acetoacetate.

3. Outline the steps in the synthesis of pentanoic acid, 2-methylpentanoic acid and methylpentanoic acid from malonic ester.

4. Give the products that will be formed from the following reactions.